

DNALONGBENCH: a benchmark suite for long-range DNA prediction tasks

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Wenduo Cheng^{1,3}, Zhenqiao Song^{2,3}, Yang Zhang^{1,3}, Shike Wang¹,
Danqing Wang², Muyu Yang¹, Lei Li²✉ & Jian Ma¹✉

Modeling long-range DNA dependencies is crucial for understanding genome structure and function across diverse biological contexts. However, effectively capturing these dependencies, which may span millions of base pairs in tasks such as three-dimensional (3D) chromatin folding prediction, remains a major challenge. A comprehensive benchmark suite for evaluating tasks that rely on long-range dependencies is notably absent. To address this gap, we introduce DNALONGBENCH, a benchmark dataset covering five key genomics tasks with long-range dependencies up to 1 million base pairs: enhancer-target gene interaction, expression quantitative trait loci, 3D genome organization, regulatory sequence activity, and transcription initiation signals. We assess DNALONGBENCH using five methods: a task-specific expert model, a convolutional neural network (CNN)-based model, and three fine-tuned DNA foundation models – HyenaDNA, Caduceus-Ph, and Caduceus-PS. We envision DNALONGBENCH as a standardized resource to enable comprehensive comparisons and rigorous evaluations of emerging DNA sequence-based deep learning models that account for long-range dependencies.

Genomic DNA sequences are the blueprint of life, guiding the development of cellular complexity. Although protein-coding DNA sequences encode diverse biochemical functions within organisms, most eukaryotic genomes consist predominantly of non-coding sequences interspersed with protein-coding regions. These non-coding sequences contain a variety of regulatory elements, such as promoters, enhancers, non-coding RNAs, and other functional elements, which orchestrate when and where genes are activated or silenced. Over the past two decades, large-scale functional genomics projects, such as ENCODE¹, have cataloged extensive collections of putative non-coding regulatory elements in the human genome. However, our understanding of how these elements regulate gene expression remains limited. A key challenge is that genomes are dynamically folded into multi-scale 3D structures within the nucleus, leading to widespread physical DNA-DNA interactions, even between regions located megabases apart^{2–4}. Determining which of these

interactions are functionally relevant across diverse biological contexts requires significant experimental effort.

The increasing availability of genomic data, such as ChIP-seq⁵, ATAC-seq⁶, and Hi-C and its derivatives⁷, has spurred the development of supervised deep learning methods that show great promise in systematically delineating sequence-to-function relationships. For example, convolutional neural networks (CNNs) and transformer-based methods have proven effective for characterizing regulatory elements^{8–11}, predicting spatial proximity between genomic loci^{12,13}, and predicting gene expressions from local sequence contexts¹⁴. Despite these advances, capturing dependencies across very long distal DNA elements remains a major computational challenge due to both the scarcity of experimental data and the difficulty of modeling long-range sequence dependencies¹⁵.

Recently, large language models have revolutionized the field of natural language processing, demonstrating remarkable capabilities

¹Ray and Stephanie Lane Computational Biology Department, School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, USA. ²Language Technologies Institute, School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, USA. ³These authors contributed equally: Wenduo Cheng, Zhenqiao Song, Yang Zhang. ✉e-mail: leili@cs.cmu.edu; jianma@cs.cmu.edu

across a wide range of applications^{16–19}. These models leverage self-supervised learning to capture complex patterns from vast amounts of unlabeled text data, followed by fine-tuning for specific tasks. Recognizing structural similarities between DNA sequences and natural language²⁰, several DNA foundation models have emerged^{21–26}. However, their utility in addressing meaningful biological questions remains a topic of debate, leaving a critical question unsolved: Could foundation models pre-trained on genomic DNA sequences offer a new paradigm shift in understanding the interactions between regulatory elements and genes? Answering this question requires robust benchmark datasets to evaluate their performance, identify limitations, and guide future improvements. Yet, most existing DNA foundation models have only been evaluated on prediction tasks involving sequences up to a few thousand base pairs, such as regulatory element identification or local gene expression prediction^{26–30}. Their potential for modeling long-range interactions in diverse biological contexts has not been well evaluated.

Benchmark datasets specifically designed to assess the ability of DNA foundation models to capture long-range dependencies remain limited. Most existing benchmarks focus on short-range tasks (spanning thousands of base pairs) and binary classification. To date, BEND²⁷ and the Genomics Long-range Benchmark (LRB)³⁰ are the only two benchmark datasets that include long-range genomic DNA prediction tasks. BEND comprises two long-range tasks: enhancer annotation and gene finding, both of which involve classifying regulatory elements. LRB, adapted from the Enformer⁴⁴ paper, curated three datasets focused on gene expression prediction and variant effects on expression. However, both are limited in scope: they emphasize regulatory element identification or gene expression prediction while overlooking other critical long-range tasks. For example, neither includes structure-related tasks requiring ultra-long sequences, such as contact map prediction or enhancer-target gene prediction. Furthermore, they lack base-pair-resolution regression tasks for quantitative assays. As a result, a comprehensive benchmark suite covering a broader range of tasks dependent on long-range DNA interactions remains absent.

Here, we introduce DNALONGBENCH, the largest collection to date of biologically meaningful long-range genomic DNA prediction tasks. DNALONGBENCH comprises five different tasks and datasets spanning critical aspects of gene regulation across multiple length scales. A comparison of existing benchmarks with DNALONGBENCH is shown in Table 1. Our contributions are threefold:

- We introduce DNALONGBENCH, a benchmark for long-range DNA prediction tasks spanning up to 1 million base pairs (bp) across five distinct tasks. To our knowledge, DNALONGBENCH is the most comprehensive benchmark specifically designed for long-range DNA prediction to date.

- We evaluate DNALONGBENCH using three representative models, demonstrating that while DNA foundation models capture long-range dependencies to some extent, expert models consistently outperform them across all tasks.
- We show that model performance varies substantially across tasks, highlighting the diverse challenges posed by DNALONGBENCH and revealing differences in task difficulty.

We envision DNALONGBENCH as a valuable resource for evaluating DNA foundation models, with particular emphasis on their ability to model long-range genomic interactions.

Results

Proposed dataset: DNALONGBENCH

The selection of suitable long-range DNA prediction tasks for DNALONGBENCH is crucial to ensure diversity, comprehensiveness, and rigor. To achieve this, we established the following criteria to guide our task selection process.

- Biological significance: Tasks should be realistic and biologically significant, addressing genomics problems important for understanding genome structure and function.
- Long-range dependencies: Tasks should require modeling long input contexts spanning hundreds of kilobase pairs or more.
- Task difficulty: Tasks should pose significant challenges for current models.
- Task diversity: Tasks should be as diverse as possible, spanning various length scales and including different task types such as classification and regression. This diversity also includes task dimensionality (1D or 2D) and granularity (binned, nucleotide-wide, or sequence-wide).

As a result, we selected five long-range DNA prediction tasks, each covering different aspects of important regulatory elements and biological processes within a cell, as illustrated in Fig. 1. An overview of our dataset is presented in Table 2. The input sequences for all tasks are provided in BED format, which lists the genome coordinates of the sequences. This format allows flexible adjustment of the flanking context without requiring reprocessing. The selected tasks are described in detail in “Methods”. Additional details on data processing, data access, and data license are provided in Supplementary Information.

Benchmarking experiments

In this section, we conduct a comprehensive performance comparison by evaluating three distinct types of models: a lightweight CNN, existing expert models that have demonstrated state-of-the-art results, and two types of recent DNA foundation models—HyeynaDNA²⁴ and

Table 1 | Comparison of existing benchmarks (Genomic Benchmarks²⁸, NT Benchmark²⁹, GUE²⁶, BEND²⁷, and LRB³⁰) for DNA prediction tasks with DNALONGBENCH

Benchmark Feature	Genomic Benchmarks	NT Benchmark	GUE	BEND	LRB	DNALONGBENCH
Has Long-range Task	x	x	x	✓	✓	✓
Longest Input (bp)	4707	600	1000	100k	192k	1M
Has Base-pair-resolution Regression Task	x	x	x	x	x	✓
Has two-dimensional Task	x	x	x	x	x	✓
Has Supervised Model Baseline	✓	x	x	✓	x	✓
Has Expert Model Baseline	x	x	x	✓	✓	✓
Has DNA Foundation Model Baseline	x	✓	✓	✓	✓	✓

While several recent benchmarks have been proposed for genomics, only BEND and LRB include tasks involving relatively long-range dependencies. In contrast, DNALONGBENCH provides the most extensive range of long-range tasks, supporting sequences up to 1 million base pairs. It also features a greater variety of task types, longer input sequences, and evaluates the performance of three representative baseline models: supervised, expert, and DNA foundation models. The supervised baseline represents fully supervised models, such as lightweight CNNs, that do not involve pre-training.

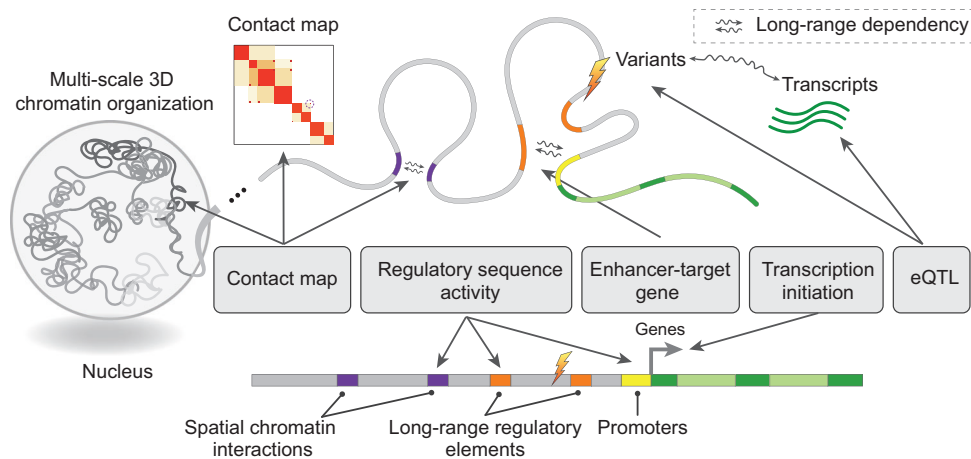


Fig. 1 | Illustration of the different categories of downstream tasks included in DNALONGBENCH. The tasks in DNALONGBENCH include contact map prediction, regulatory sequence activity, enhancer-target gene linking, transcription initiation, and eQTL mapping, covering key layers of chromatin architecture and gene regulation that depend on long-range DNA interactions.

Table 2 | Overview of the tasks included in DNALONGBENCH

LR Tasks	LR Type	Input Length	Output Shape	# Samples	Metric
Enhancer-target Gene	Binary Classification	450,000	1	2602	AUROC
eQTL	Binary Classification	450,000	1	31,282	AUROC
Contact Map	Binned (2048 bp) 2D Regression	1,048,576	99681	7840	SCC&PCC
Regulatory Sequence Activity	Binned (128 bp) 1D Regression	196,608	Human: (896, 5313) Mouse: (896, 1643)	Human: 38,171 Mouse: 33,521	PCC
Transcription Initiation Signal	Nucleotide-wise 1D Regression	100,000	(100,000, 10)	100,000*	PCC

*1D" and "2D" denote one-dimensional and two-dimensional, respectively. Nucleotide-wise tasks involve predicting a sequence of labels, each corresponding to individual nucleotides in the input. Sequence-wise tasks require classifying the entire input sequence. In binned tasks, multiple nucleotides are grouped into bins and share a common label. *: The data for this task consists of sequences sampled from whole genomes, with 100,000 samples used for training our baselines. AUROC area under the receiver operating characteristic curve, PCC Pearson correlation coefficient, SCC stratum-adjusted correlation coefficient.

Caduceus²⁵—distinguished by their support for reverse complement DNA during the training process.

Representative models. We explore the performance of the following three types of models:

- (1) CNN: We evaluate the lightweight convolutional neural network³¹, known for its simplicity and robust performance in various DNA-related tasks. For classification tasks, we trained a three-layer CNN using cross-entropy loss. For contact map prediction, we designed a CNN combining 1D and 2D convolutional layers, trained with mean squared error (MSE) loss. For the regulatory sequence activity and transcription initiation signal prediction tasks, we used CNNs trained with Poisson loss and MSE loss, respectively.
- (2) Expert Model: We assess the current state-of-the-art specialized models for each specific long-range DNA prediction task, collectively referred to as the expert model. Specifically, we use:
 - The Activity-by-Contact (ABC) model³² for the enhancer-target gene prediction task.
 - Enformer¹⁴ for the eQTL prediction task and regulatory sequence activity prediction task.
 - Akita¹² for the contact map prediction task.
 - Puffin-D³³ for the transcription initiation signal prediction task.
- (3) DNA Foundation Model: We selected three long-range DNA foundation models—HyenaDNA (medium-450k)²⁴ and Caduceus (Ph and PS)²⁵—for evaluation, as they are published works specifically designed for long-range DNA prediction tasks. For the eQTL task, we extracted last-layer hidden representations

Table 3 | AUROC and AUPRC for the enhancer-target gene prediction (ETGP) task

Models	Enhancer-Target Gene Prediction	
	AUROC	AUPRC
Expert Model	0.926	0.407
CNN	0.797	0.310
HyenaDNA	0.828	0.249
Caduceus-Ph	0.826	0.380
Caduceus-PS	0.821	0.376

The highest scores are highlighted in bold.

from both the reference and allele sequences, averaged and concatenated them, and applied a binary classification layer to predict whether the variant was positive. For the remaining tasks, we fed the DNA sequences into the DNA foundation model to obtain feature vectors, then applied linear layers to predict logits at different resolutions.

More detailed model implementations for each task are provided in the Supplementary Information.

Expert models achieve the highest scores on all tasks. We summarize our evaluation results into five tables for each task, respectively, as shown in Tables 3–7. For instance, Table 3 shows the AUROC and AUPR metrics for the enhancer-target gene prediction task, with additional results in Table S1 and Table S2. Table 4 and Table S3 summarize stratum-adjusted correlation coefficient and

Table 4 | Stratum-adjusted correlation coefficient for the contact map prediction (CMP) task

Models	Contact Map Prediction					
	HFF	H1hESC	GM12878	IMR90	HCT116	Avg
Expert Model	0.258	0.247	0.227	0.210	0.210	0.230
CNN	0.025	0.024	0.010	0.013	0.001	0.015
HyenaDNA	0.139	0.122	0.099	0.097	0.118	0.115
Caduceus-Ph	0.153	0.130	0.101	0.138	0.145	0.133
Caduceus-PS	0.142	0.123	0.097	0.132	0.139	0.127

K562, HFF, H1hESC, GM12878, IMR90, and HCT116 represent different human cell types. The highest scores are highlighted in bold. “Avg” means the average score across different cell types.

Table 5 | Pearson correlation scores for the regulatory sequence activity prediction (RSAP) task

Models	Regulatory Sequence Activity Prediction		
	Human	Mouse	Avg
Expert Model	0.669	0.479	0.574
CNN	0.538	0.323	0.431
HyenaDNA	0.298	0.396	0.347
Caduceus-Ph	0.301	0.400	0.349
Caduceus-PS	0.301	0.387	0.344

The highest scores are highlighted in bold.

Table 6 | Pearson correlation scores for the transcription initiation signal prediction (TISP) task

Models	Transcription Initiation Signal Prediction					
	FC	EC	ER	GC	PC	Avg
Expert Model	0.808	0.710	0.749	0.624	0.774	0.733
CNN	0.029	0.038	0.043	0.037	0.066	0.042
HyenaDNA	0.138	0.124	0.118	0.112	0.168	0.132
Caduceus-Ph	0.114	0.088	0.088	0.097	0.154	0.109
Caduceus-PS	0.113	0.088	0.090	0.102	0.156	0.108

FANTOM CAGE (FC), ENCODE CAGE (EC), ENCODE RAMPAGE (ER), GRO-cap (GC), and PRO-cap (PC) represent the experimental techniques used in the TISP task. “Avg” means the average scores. The highest scores are highlighted in bold.

Table 7 | AUROC scores for the expression quantitative trait loci prediction (eQTLp) task across different cell types

Models	Expression Quantitative Trait Loci Prediction									
	CCF	WB	Thyroid	SNSSES	SSELL	MS	NT	AT	AS	Avg
Expert Model	0.639	0.689	0.612	0.710	0.700	0.621	0.683	0.741	0.736	0.681
CNN	0.547	0.577	0.487	0.499	0.499	0.502	0.516	0.576	0.551	0.528
HyenaDNA	0.584	0.512	0.529	0.471	0.544	0.487	0.511	0.479	0.513	0.514
Caduceus-Ph	0.597	0.594	0.527	0.586	0.574	0.538	0.588	0.547	0.541	0.565
Caduceus-PS	0.549	0.542	0.547	0.529	0.541	0.523	0.552	0.536	0.519	0.537

The cell types are abbreviated as follows: CCF for Cells_Cultured_fibroblasts, WB for Whole_Blood, SNSSES for Skin_Not_Sun_Exposed_Suprapubic, SSELL for Skin_Sun_Exposed_Lower_leg, MS for Muscle_Skeletal, NT for Nerve Tibial, AT for Artery_Tibial, and AS for Adipose_Subcutaneous. Notably, Expert Model achieves the best performance across all cell types. The highest scores are highlighted in bold. “Avg” represents the average score across all cell types.

Pearson correlation for the contact map prediction task for five primary cell lines, with results for four additional cell types shown in Table S4. Figure 2 and Fig. S1 demonstrate examples of contact maps predicted by different methods and ground truth. Table 7 and Table S5 show the AUROC and AUPRC for the eQTL prediction task. In general, we observed that highly parameterized and specialized expert models consistently outperform DNA foundation models. Notably, the advantages of these expert models appear greater in regression tasks such as contact map prediction and transcription initiation signal prediction than in classification task (e.g., enhancer-target gene prediction). For instance, the expert model Puffin achieves an average score of 0.733 on the transcription initiation signal prediction task (TISP), significantly surpassing CNN (0.042), HyenaDNA (0.132), Caduceus-Ph (0.109), and Caduceus-PS (0.108).

This disparity may stem from the challenge posed by multi-channel regression on long DNA contexts, which makes fine-tuning of DNA foundation models less stable and less capable of capturing sparse real-valued signals. We acknowledge that these expert models are specially designed for their respective tasks, and that some—such as Enformer—have more parameters than HyenaDNA and Caduceus, thereby serving as both strong baselines and potential upper bounds for all tested models. Overall, these observations confirm the Expert Model’s superior ability to capture long-range dependencies, a capability where CNN falls short and DNA foundation models demonstrate reasonable performance in certain tasks.

The contact map prediction presents greater challenges. Unlike the other four tasks, where the Expert Model or DNA foundation models achieve reasonable performance, the contact map prediction task proves significantly more difficult. The highest stratum-adjusted correlation coefficient achieved in this task is 0.233 by the Expert Model (Akita), indicating only a moderate positive correlation. Although contact map prediction is crucial for understanding 3D genome structure, it has received less attention in previous benchmarks, which focused primarily on 1D prediction tasks. This highlights both the difficulty of modeling long-range genomic interactions and the varying levels of complexity across tasks in DNALONGBENCH.

Longer contexts improve model performance. To investigate whether the tasks in our benchmark require long contexts to achieve strong results, we performed ablation studies. This was done by either using varying context lengths or shuffling the central portion of the input sequence, with results reported in Tables S6–S11. For instance, for the contact map prediction task, we chose Caduceus-Ph for ablation since it showed the highest SCC among the DNA foundation models, and evaluated its performance with input sizes of 409,600, 307,200, and 204,800 bp, corresponding to 200, 150, and 100 bins, respectively. Our results show that model per-

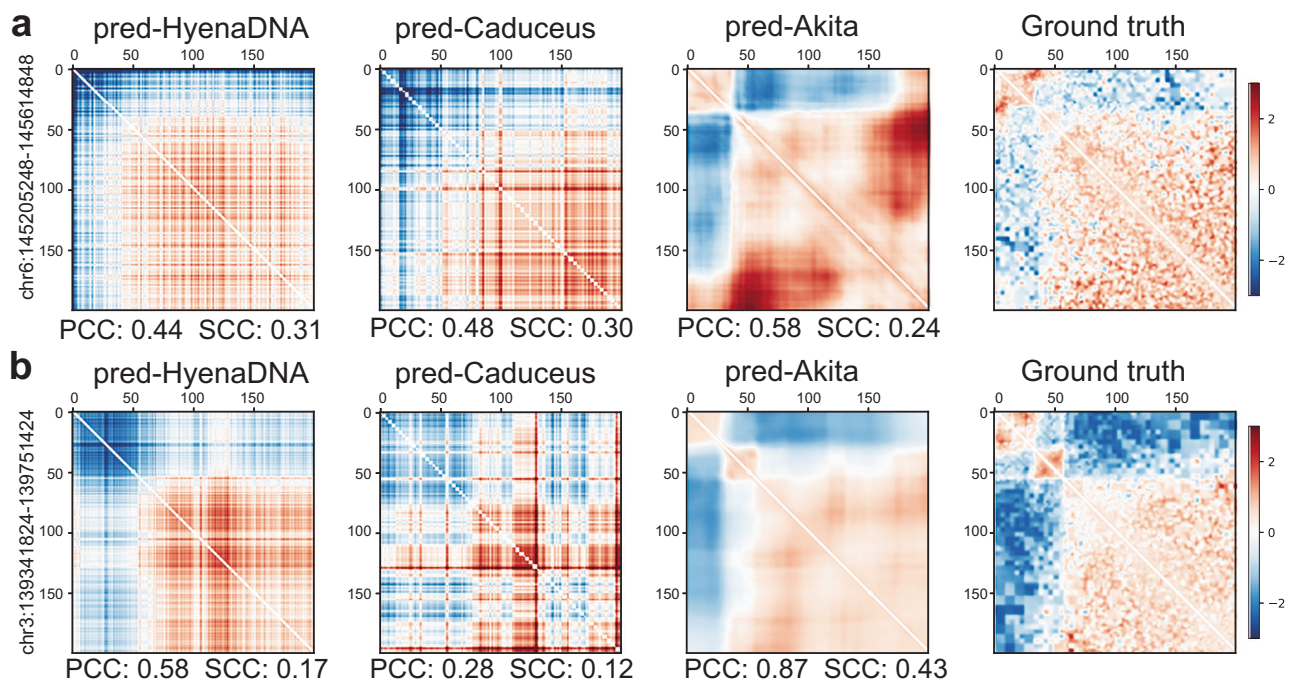


Fig. 2 | Comparisons of HyenaDNA, Caduceus (Ph), and the Expert Model (Akita) on the 2D contact map prediction task across 409,600 bp with a bin size of 2048 bp. The columns show contact maps predicted by HyenaDNA, Caduceus, and Akita model, alongside the ground truth contact map for two genomic regions: **a** chr6:145,205,248–145,614,848 and **b** chr3:139,341,824–139,751,424. Colors

represent the intensity of contact frequency between paired loci. Pearson correlation coefficient (PCC) and stratum-adjusted correlation coefficient (SCC) metrics are shown beneath each contact map to indicate prediction performance relative to the ground truth. Source data are provided as a Source Data file.

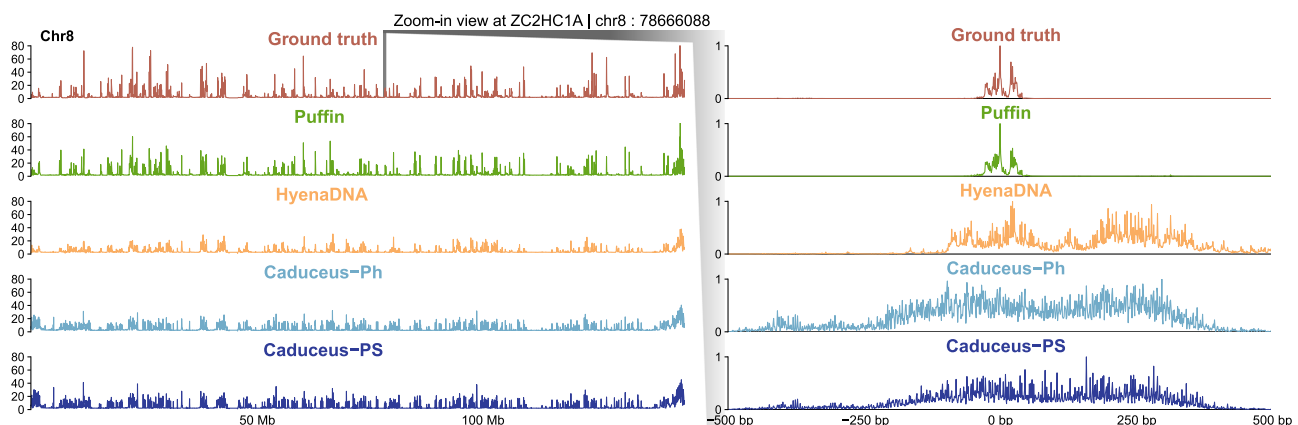


Fig. 3 | Comparisons of HyenaDNA, Caduceus-Ph, Caduceus-PS, and Expert Model (Puffin-D) on the transcription initiation signal prediction task of chromosome 8. The genomic track on the left displays the ground truth signals (top) alongside predictions from Puffin-D, HyenaDNA, and the two Caduceus

models. The X-axis represents genomic coordinates, while the Y-axis indicates signal density. A zoomed-in view of a 1000 bp region centered at the TSS of the gene *ZC2HC1A* is shown on the right. Source data are provided as a Source Data file.

formance increases as context length increases. Similar trends are observed in the other tasks as well, suggesting the model benefits from longer contexts.

Further analysis of DNALONGBENCH evaluations

In this section, we provide further analysis to gain insight into how long-range dependencies are captured in our proposed DNALONGBENCH.

Case Study: Can long-range dependency be captured? To intuitively demonstrate the presence of extensive long-range dependencies across millions of base pairs and their capture by machine learning methods, we present two examples in Fig. 2 and more examples in

Fig. S1. Specifically, in Fig. 2a, b, we visualize the contact maps predicted by HyenaDNA, Caduceus-Ph, and the Expert Model (Akita), alongside the ground truth contact maps for two genomic regions spanning around 400 kb. From these contact maps, we observe the presence of large-scale domains (e.g., blocks in the contact map) and long-range interactions (e.g., off-diagonal dots in the contact map) spanning over 300 kb. Notably, the contact maps predicted by Akita align more closely with the ground truth, confirming its superior ability to capture long-range interactions. In contrast, DNA foundation models show a limited capacity to predict domain structures. This is particularly evident in Fig. 2b, where only Akita accurately predicts the three blocks. These examples highlight DNALONGBENCH's value in evaluating models for capturing long-range genome structure and

function, and provide a foundation for future developments in DNA foundation models.

Base pair-resolution prediction of transcription initiation signal. We visualized the transcription initiation signals predicted by different models for one of the test chromosomes, chromosome 8 (Fig. 3). Predictions from the Expert model Puffin-D closely align with the ground truth, accurately capturing peaks in transcription initiation signal intensity across both large and small genomic regions. In contrast, DNA foundation models tend to underpredict signal intensities or miss certain peaks. In the zoomed-in view (right side of the figure), Puffin-D continues to align well with the ground truth, demonstrating strong performance even at high resolution. By contrast, the DNA foundation models show less precise and broader signals. These findings suggest that base pair-resolution regression tasks remain challenging for current DNA foundation models.

Discussion

In this paper, we introduce DNALONGBENCH, a benchmark suite comprising five important genomics tasks involving long-range dependencies: enhancer-target gene interaction, eQTL, 3D genome organization, regulatory sequence activity, and transcription initiation signals. We evaluated three baseline methods: a task-specific expert model, a fully supervised CNN-based model, and three fine-tuned DNA foundation models, HyenaDNA, Caduceus-Ph, and Caduceus-PS. The benchmarking results consistently showed that expert models achieved the highest scores across all tasks. Additionally, our analysis revealed that long-range dependencies could be captured across hundreds of thousands of base pairs, underscoring the importance of context length for downstream performance. However, the results also highlight that current DNA foundation models are less effective than expert models in capturing long-range dependencies. It is important to note that each expert model was specifically designed and trained for its respective task. In contrast, DNA foundation models are intended as a “one-to-all” general-purpose solution across diverse applications. Consequently, simple fine-tuning may not be sufficient to outperform these highly specialized expert architectures. There remains substantial room to improve foundation models through novel architectural designs, advanced fine-tuning strategies, and task-specific training objectives. Nevertheless, we believe that DNALONGBENCH will serve as a valuable resource for enabling comprehensive comparisons and rigorous evaluations of emerging DNA sequence-based deep learning models that account for long-range dependencies.

One limitation of this study is the exclusion of transformer-based DNA foundation models, such as DNABERT-1, DNABERT-2, and Nucleotide Transformer, due to the computational challenges posed by training them on long-range tasks. The quadratic cost of the self-attention mechanism renders such tasks infeasible for these models. Exploring strategies to extend the context length of transformer-based models and effectively fine-tune them for long-range tasks remains an important avenue for future research, albeit beyond the scope of this study.

Methods

Benchmark dataset: enhancer-target gene prediction

In eukaryotic cells, enhancers play a key role in gene regulation by forming enhancer-promoter interactions that activate the transcription of target genes, even those located up to several megabases away³⁴. However, the detailed mechanism by which sequence information encodes enhancer–promoter interactions remains poorly understood. Predictive methods that incorporate the entire sequence between enhancers and promoters as input could not only improve prediction performance but also help identify the sequence determinants driving these interactions. To this end, we formulated a task to

predict true enhancer–promoter interactions from a list of putative candidates based on the DNA sequence.

We collected experimentally verified enhancer–promoter interactions in K562 cells from three studies^{32,35,36}. Using CRISPR-mediated perturbation techniques, the authors perturbed thousands of candidate sequences, quantified their effects on gene expression, and identified both positive and negative enhancer–promoter interactions. We filtered this data by retaining enhancer–promoter pair candidates within 450 kb of the gene transcription start site (TSS) and applied additional filtering criteria. Model performance was evaluated using AUROC. We compared models that rely solely on sequence information with the expert model, the Activity-by-Contact (ABC) model³², which incorporates DNase-seq, H3K27ac ChIP-seq data, and a Hi-C matrix to prioritize true enhancer–promoter interactions. It should be noted that the ABC model has inherent advantages over sequence-only models due to its more comprehensive input data types. The primary motivation here is to compare sequence-only models and understand their strengths and limitations.

Benchmark dataset: 3D chromatin contact map prediction

Chromosomes are folded in a well-organized manner within the cell nucleus, affecting various critical cellular functions such as gene transcription and DNA replication^{37,38}. Developing prediction models that connect 1D DNA sequences with 2D contact maps enables the identification of key sequence determinants of 3D chromatin folding, providing valuable insights into the underlying mechanisms of genome organization^{4,39}. We formulated a 3D chromatin contact map prediction task, defined as a 2D regression task to predict pairwise chromatin interactions between every pair of genomic loci within a given context window.

These contact frequencies are expressed as 2D contact maps derived from genomic mapping data such as Hi-C and Micro-C⁴. We used the processed data from Akita¹², which includes chromatin interaction data from five cell lines: HFF, H1-hESC, GM12878, IMR-90, and HCT116. To increase the number of cell types, we curated and processed additional Hi-C data for four cell lines: HAP1, HeLa, HepG2, and K562 from the 4DN data portal⁴⁰, following the same data processing steps as in the Akita model. Each input sequence, spanning 1 million base pairs (Mbp), is divided into 512 genomic bins at a resolution of 2 kb per bin. For the final prediction, 32 genomic bins are cropped from each side, resulting in a contact map of 448 × 448. Since the contact map is symmetric, predictions are made only for the upper triangular region, with a diagonal offset of 2. The human genome was divided into non-overlapping virtual contigs and randomly assigned to training, validation, and testing sets with an 8:1:1 ratio. The dataset contains 7008 training sequences, 419 validation sequences, and 413 test sequences. Model performance on the held-out test set was evaluated using the stratum-adjusted correlation coefficient (SCC) and the Pearson correlation coefficient (PCC).

Benchmark dataset: regulatory sequence activity prediction

Cell type-specific regulatory activities are encoded by the compositions and interactions of functional DNA segments, such as promoters, enhancers, and insulators, which can regulate genes from distant genomic locations. Predicting functional signals directly from DNA sequences spanning large genomic distances could help identify distal regulatory elements and uncover key sequence features that enable long-range gene regulation. For this task, we compiled human and mouse genomic tracks from the Enformer paper¹⁴. The goal of this task is to predict thousands of epigenomic profiles directly from DNA sequence spanning up to 100 kb. We formulated the task as a multitask regression problem aimed at predicting epigenetic and transcriptional signals from long DNA sequences alone.

The dataset includes experimentally determined regulatory activity signal tracks and corresponding DNA sequences from human and mouse genomes. Each input DNA sequence spans 196,608 bp, centered on the TSS of protein-coding genes. Each input sequence consists of a core region and flanking regions. The core sequence is 114,688 bp in length, corresponding to 896 bins at a resolution of 128 bp per bin. The target labels consist of 5313 human tracks and 1643 mouse tracks measuring epigenomic marks. The dataset contains 38,171 human sequences and 33,521 mouse sequences. For the human genome, the data is split into 34,021 training, 2213 validation, and 1937 test sequences. For the mouse genome, the dataset is split into 29,295 training, 2209 validation, and 2017 test sequences. Model performance was evaluated using the Pearson correlation coefficient, calculated by comparing predicted and target signal tracks. Specifically, the Pearson correlation coefficients were computed for each sample across all positions and tracks, and the mean was taken across all samples in the test set.

Benchmark dataset: eQTL prediction

Expression quantitative trait loci (eQTL) are nucleotide variants that affect the expression of one or more genes. Deep learning-based approaches for predicting gene expression from DNA sequences have gained increasing popularity. One practical application of these methods is the identification and interpretation of eQTLs, a traditionally labor-intensive and time-consuming process when relying on genome-wide association studies. We designed an eQTL prediction task to provide an efficient approach for evaluating eQTLs, where the goal is to predict whether a nucleotide variant modulates the expression of a target gene using DNA sequence alone.

We adapted the eQTL dataset used in Enformer¹⁴. Positive SNPs were identified using the statistical fine-mapping tool SuSiE⁴¹. The original dataset includes positive and matched negative variants across 48 tissues¹⁴. For this study, we selected the top nine tissues based on the number of variants. Within these tissues, eQTL-gene pairs were filtered to retain eQTL candidate loci within 450 kb of the gene TSS. Genes with fewer than two positive pairs, two negative pairs, or five combined pairs were excluded. The sequences between variants and promoters were extracted, extending 3 kb downstream of the gene TSS. To reduce bias caused by putative eQTLs within the interval between an eQTL candidate and the gene promoter pair, we masked the sequences of all variants within each variant-promoter pair. The dataset was randomly split into training, validation, and test sets using a stratified sampling approach with an 8:1:1 ratio. To ensure robustness, at least one positive pair and one negative pair were included in both the training and validation sets. Model performance was evaluated using AUROC.

Benchmark dataset: transcription initiation signal prediction

Promoters are specialized DNA sequences at the TSS of genes that support the assembly of transcription machinery and transcription initiation⁴². Each promoter exhibits a unique profile of transcription initiation signals, which may reflect the mechanisms underlying transcription initiation. Solving the machine learning task of predicting these profiles from promoter sequences provides insights into sequence-based regulation of transcription initiation³³. Using long sequences as input and improving the information flow between distal elements could enhance the predictive accuracy of transcription initiation signal prediction. We include a task in DNALONGBENCH aimed at predicting transcription initiation signal profiles from DNA sequences. Specifically, the task predicts transcription initiation signals on both strands for five experimental techniques: FANTOM CAGE, ENCODE CAGE, ENCODE RAMPAGE, GRO-cap, and PRO-cap³³. Unlike the regulatory sequence activity prediction task, which predicts sequence coverage at 128 bp genomic

bins, this task requires predictions at base-pair resolution, making it significantly more challenging.

We used processed labeled data from the Puffin model³³. Predictions were generated for entire test chromosomes (chr8 and chr9) using a sliding window with a step size of 50 kb, with the center 50 kb of each 100 kb prediction being evaluated. Regions within 1 kb of unknown bases or within 25 kb of chromosome ends were excluded. Model performance was evaluated using the Pearson correlation coefficient.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The benchmark datasets have been deposited to Harvard DataVerse under the following DOI link: 1. Regulatory Sequence Activity Prediction Data: <https://doi.org/10.7910/DVN/MNUEZR>; 2. Transcription Initiation Signal Prediction Data: <https://doi.org/10.7910/DVN/VXQKWO>; 3. Enhancer-Target Gene Prediction Data: <https://doi.org/10.7910/DVN/CTEQXX>; 4. 3D Chromatin Contact Map Prediction Data: <https://doi.org/10.7910/DVN/AZM25S>; 5. Expression Quantitative Trait Loci (eQTL) Prediction Data: <https://doi.org/10.7910/DVN/YUP2G5>. The enhancer-target gene dataset was obtained from CRISPRi-based screening data from three studies^{32,35,36}. The contact map prediction data were derived from the previous Akita¹² paper at <https://github.com/calico/basenji/tree/master/manuscripts/akita> and four additional in situ Hi-C datasets (4D Nucleome Data Portal accession IDs: 4DNFIWGGYEW2, 4DNFI65WJKMT, 4DNFIQ4G74OW, 4DNFI2R1W3YW). The eQTL and regulatory sequence activity data were obtained from the Basenji⁴³ paper, which was previously used by the Basenji2⁴⁴ and Enformer¹⁴ models, available at https://console.cloud.google.com/storage/browser/basenji_barnyard/data. The transcriptional initiation signal prediction data were obtained from Zenodo at 10.5281/zenodo.7954971⁴⁵. Source data are provided with this paper.

Code availability

The source code is available on GitHub at <https://github.com/macompbio/DNALONGBENCH>, under the BSD-3-Clause license. The specific version of the code associated with this publication is archived in Zenodo and is accessible via <https://doi.org/10.5281/zenodo.17179568>⁴⁶.

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Author contributions

Conceptualization, W.C., Z.S., Y.Z., L.L., J.M.; Methodology, W.C., Z.S., Y.Z.; Software, W.C., Z.S., Y.Z., S.W., D.W., M.Y.; Investigation, W.C., Z.S., Y.Z., S.W., D.W., M.Y., L.L., J.M.; Writing, W.C., Z.S., Y.Z., J.M.; Funding Acquisition, L.L., J.M.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Lei Li or Jian Ma.

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